Mohammed Elmahfoudi

Software Engineer

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Professional Summary

Software Engineer with a strong foundation in systems programming and backend development, building everything from raw HTTP servers in C++ to production APIs and containerised infrastructure. Proficient in Python for backend development and data engineering workflows. Currently advancing through 1337 Coding School (42 Network), and real world projects spanning RBAC systems, DevOps pipelines, and full stack platforms.

Experience

Backend Developer | *SHBM INF Platform — UM6P* **Jan 2025 – Present**

Next.js · Node.js · REST APIs · RBAC · PostgreSQL · Remote

- Architected RESTful API services using Next.js API routes, powering student and administrative workflows for a university scale platform.
- Engineered a JWT based RBAC system securing multi-tenant access across student, faculty, and admin roles; optimised DB queries and API response times for concurrent user loads.

Entrepreneurship Intern | *UM6P Explorer Program (MIT Sandbox)* **Jun – Sep 2025**

Product Development · Market Validation · Agile · Benguerir

- Validated a healthcare accessibility prototype with 20+ users; delivered working MVP under UM6P & MIT Sandbox mentorship using lean startup methodology to achieve confirmed product market fit.

Technical Projects

Inception — Containerized Infrastructure | *42 Network* *Docker · NGINX · MariaDB · WordPress*

- Designed and deployed a secure multi service infrastructure using Docker Compose, orchestrating isolated containers for NGINX, MariaDB, and WordPress.
- Implemented persistent storage, custom networking, and service configuration in a Linux environment.
- Strengthened skills in system administration, virtualization, and infrastructure automation.

Webserv — HTTP Server from Scratch | *42 Network* *C++ · HTTP/1.1 · Sockets · Linux · I/O Multiplexing*

- Developed a fully functional HTTP/1.1 web server in C++ without external frameworks.
- Implemented request parsing, response generation, routing, static file serving, and CGI execution.
- Used NON-blocking I/O and event driven architecture to handle multiple client connections efficiently.

Ft_Transcendence — Full-Stack Web Platform | *42 Network* *Next.js · FastAPI · PostgreSQL · Docker*

- Built a full stack web platform with role based auth (candidate & recruiter), JWT session handling, and protected routing via Next.js middleware.
- Engineered an AI powered recommendation engine exposed via REST API, with criteria based filtering and real time result rendering on the frontend.
- Designed a multi role recruiter flow for offer management including creation, listing, and active/inactive toggling with live UI updates.

Technical Skills

Languages: Python, JavaScript, TypeScript, C, C++, SQL, Bash

Backend: FastAPI, NestJS, Next.js (API Routes), REST APIs

DevOps: Docker, GitHub Actions, CI/CD, Linux, Nginx

Databases: MySQL, PostgreSQL, MongoDB

Frontend: React.js, HTML5, CSS3

Tools: Git, GitHub, Postman, Figma, VS Code, Docker Hub

Education

1337 Coding School (42 Network) | Software Engineering Program **2024 – Present**

Ecole Supérieure de Technologie, Kénitra | DUT in Computer Science **2022 – 2024**

ALX Africa | Data Engineering Program (Remote) — ETL, data modelling, cloud architecture **2025 – Present**